Example 2 - Problem Statement:

Given a system of two particles moving in a 2D plane:

- **Particle A_1:**
  - Mass (m_1): $1\ \mathrm{kg}$
  - Velocity ($\vec{V}_1$): $(2\vec{i} - 3\vec{j})\ \mathrm{m/s}$
- **Particle A_2:**
  - Mass (m_2): $200\ \mathrm{g}$
  - Velocity ($\vec{V}_2$): $(-3\vec{i} + 4\vec{j})\ \mathrm{m/s}$

Determine $\vec{V}_G$ (the velocity vector of the center of mass) and V_G (the magnitude of that velocity).

Step-by-Step Solution:

This problem follows the same principles as Example 1, but now we must handle the $\vec{i}$ (x-axis) and $\vec{j}$ (y-axis) components separately.

1. Standardize Units (SI Units): First, convert all masses to kilograms.

- $m_1 = 1\ \mathrm{kg}$
- $m_2 = 200\ \mathrm{g} = 0.2\ \mathrm{kg}$

2. Calculate the Total Linear Momentum of the System ($\vec{P}_S$): Using **Section B** from your notes, we sum the individual momentum vectors:

$$\vec{P}_S = \vec{P}_1 + \vec{P}_2$$

$$\vec{P}_S = m_1\vec{V}_1 + m_2\vec{V}_2$$

Substitute the values:

$$\vec{P}_S = 1 \cdot (2\vec{i} - 3\vec{j}) + 0.2 \cdot (-3\vec{i} + 4\vec{j})$$

Distribute the scalar masses:

$$\vec{P}_S = (2\vec{i} - 3\vec{j}) + (-0.6\vec{i} + 0.8\vec{j})$$

Group the $\vec{i}$ and $\vec{j}$ components:

$$\vec{P}_S = (2 - 0.6)\vec{i} + (-3 + 0.8)\vec{j}$$

$$\vec{P}_S = 1.4\vec{i} - 2.2\vec{j}\ (\mathrm{kg \cdot m/s})$$

3. Calculate the Total Mass of the System (M_S):

$$M_S = m_1 + m_2$$

$$M_S = 1 \text{ kg} + 0.2 \text{ kg}$$

$$M_S = 1.2 \text{ kg}$$

4. Determine the Velocity Vector of the Center of Mass ($\vec{V}_G$): Using **Section C** from your notes ($\vec{P}_S = M_S \cdot \vec{V}_G$), we solve for $\vec{V}_G$:

$$\vec{V}_G = \frac{\vec{P}_S}{M_S}$$

$$\vec{V}_G = \frac{1.4\vec{i} - 2.2\vec{j}}{1.2}$$

$$\vec{V}_G = \left(\frac{1.4}{1.2}\right)\vec{i} - \left(\frac{2.2}{1.2}\right)\vec{j}$$

$$\vec{V}_G \approx 1.17\vec{i} - 1.83\vec{j} \text{ m/s}$$

5. Determine the Magnitude of the Velocity (V_G): To find the magnitude (speed) of the center of mass, we use the Pythagorean theorem on its velocity components ($V_x \approx 1.17$ and $V_y \approx -1.83$).

$$V_G = ||\vec{V}_G|| = \sqrt{(V_x)^2 + (V_y)^2}$$

$$V_G = \sqrt{(1.17)^2 + (-1.83)^2}$$

$$V_G = \sqrt{1.3689 + 3.3489}$$

$$V_G = \sqrt{4.7178}$$

$$V_G \approx 2.17 \text{ m/s}$$

Final Answer:

- The velocity vector of the center of mass is: $\vec{V}_G \approx 1.17\vec{i} - 1.83\vec{j} \text{ m/s}$
- The magnitude of the velocity (speed) of the center of mass is: $V_G \approx 2.17 \text{ m/s}$